

Lesson Objective

Use visual models to add and subtract two fractions with the same units.

Materials: Personal white board

1. Subtract fractions with like units using a number line.

Solve $5/6 - 3/6$ using the number line.



Answer: 5 sixths - 3 sixths = 2 sixths; $5/6 - 3/6 = 2/6$

Remind students that fractions with the same denominator share a unit, just like ones, apples, or meters.

Ask students to say the number sentence in unit form: 5 sixths minus 3 sixths.

Guide students through partitioning the number line into sixths between 0 and 1 and plotting a point at $5/6$.

Have students count backward by sixths, drawing an arrow from $5/6$ back 3 sixths, and label where they land.

Ask students to write the subtraction sentence in standard form using fractions and confirm $5/6 - 3/6 = 2/6$.



2. Add fractions with like units using a number line, and rename the sum greater than 1 as a mixed number using a number bond.

Solve $5/4 + 2/4$ using the number line. Then use a number bond to rename the sum as a mixed number.



Answer: 5 fourths + 2 fourths = 7 fourths; $5/4 + 2/4 = 7/4$; $7/4 = 1$ and $3/4$

Ask students to say the addition sentence in unit form: 5 fourths plus 2 fourths.

Have students partition the number line into fourths between 0 and 2, plot a point at $5/4$, and count forward 2 fourths to land on $7/4$.

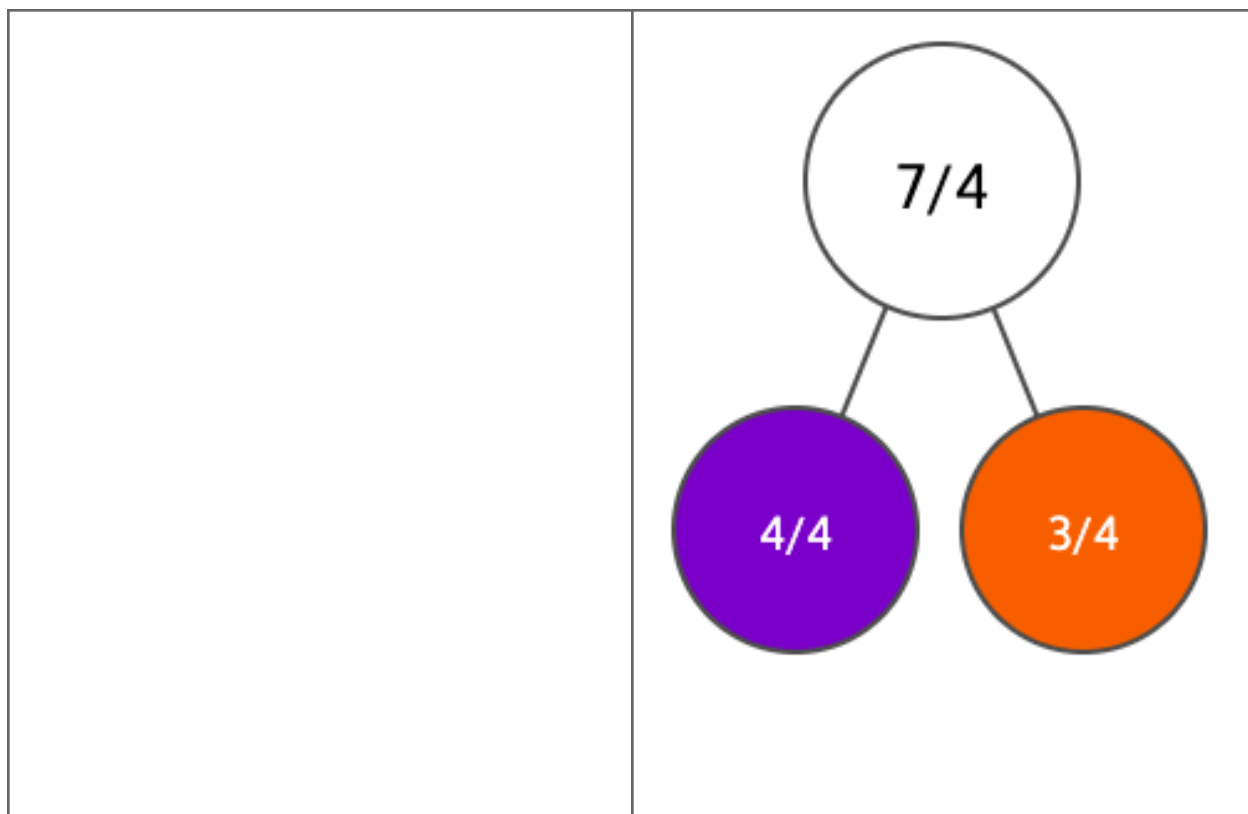
Ask students to write the addition sentence in standard form using fractions.

Ask students how many fourths are in 1 whole, then have them draw a number bond decomposing $7/4$ into $4/4$ and $3/4$.

Point out that $4/4$ equals 1 whole, so $7/4$ can be renamed as the mixed number 1 and $3/4$.

Confirm that $7/4$ and 1 and $3/4$ name the same amount.



**Monitor For**

- Do students partition the number line into equal parts that match the denominator before plotting the starting fraction?
- Do students count fractional units (sixths, fourths) when moving along the number line, rather than counting whole-number tick marks?
- Do students write the subtraction or addition sentence in both unit form and standard fraction form?

Instructional Moves

- Have students say each sentence in unit form before writing it in standard form (e.g., "5 sixths minus 3 sixths equals 2 sixths") to reinforce that fractions count like any other unit.
- If a student counts tick marks instead of intervals on the number line, have them touch and name each fractional unit as they move (1 sixth, 2 sixths, 3 sixths).

• Do students recognize that $\frac{4}{4}$ equals 1 whole when decomposing $\frac{7}{4}$ in the number bond?

• In Problem 2, before drawing the number bond, ask students how many fourths make 1 whole so they can pull $\frac{4}{4}$ out of $\frac{7}{4}$ cleanly.

• If a student writes $\frac{7}{4}$ and stops, prompt them to circle the $\frac{4}{4}$ in the number bond and ask, "What number is that the same as?" to bridge to the mixed number.

Reflection Questions

• How is subtracting $\frac{5}{6} - \frac{3}{6}$ like subtracting $5 - 3$? *Answer: When the units are the same, we can subtract the numerators just like subtracting whole numbers. 5 sixths minus 3 sixths is 2 sixths, the same way 5 ones minus 3 ones is 2 ones.*

• How did the number line help you add $\frac{5}{4} + \frac{2}{4}$? *Answer: I plotted $\frac{5}{4}$ and counted forward 2 fourths along the number line to land on $\frac{7}{4}$.*

• How did the number bond help you rename $\frac{7}{4}$ as a mixed number? *Answer: I decomposed $\frac{7}{4}$ into $\frac{4}{4}$ and $\frac{3}{4}$. Since $\frac{4}{4}$ equals 1 whole, I could rename $\frac{7}{4}$ as 1 and $\frac{3}{4}$.*

• Are $\frac{7}{4}$ and 1 and $\frac{3}{4}$ equal? How do you know? *Answer: Yes. They are two different ways to name the same amount. The number bond shows that $\frac{7}{4}$ is made of 1 whole and $\frac{3}{4}$.*